An AI-assisted investigation of a Spin(10) × global $U(1)_{PQ}$ gauge/scalar candidate

Construction, bounded results, and failed tests

Solo research discussion note, prepared with AI assistance. Version 3.

Abstract

This note specifies a nonsupersymmetric Spin(10) gauge/scalar construction with global PQ selection rules and reports bounded calculations. Earlier locally positive scalar points include a failed Higgs quartic tuning and a positive-quartic point subsequently rejected by a tree perturbativity screen. A different exploratory point has positive heavy masses, a positive relaxed Higgs quartic, a full-field quartic-positivity certificate and a complete scalar-contact bound below 1/2. Its formal one-loop singlet pole correction, however, exceeds its tree squared mass. Gauge unification and complete quantum consistency remain unresolved. Local anomaly traces and conditional source identities add necessary checks, not a viability theorem. This is a discussion draft based on recorded evidence, with no new scientific acceptance decision.

1. Assumptions and fields

Work in four spacetime dimensions with gauge group Spin(10) and global PQ selection rules. The simply connected group permits the fermionic spinor 16; local tensor calculations use the $\mathfrak{so}(10)$ action. A possible quotient by common gauge/PQ centers has not been fixed, although the ordinary-field center identification has now been calculated (Section 5). No supersymmetry, explicit soft PQ violation, or additional scalar representation is assumed.

Field	Representation/type	q_{PQ}	Real scalar coordinates
Φ_{ij}	real symmetric traceless 54	0	54
Σ_{ijklm}	complex self-dual 126	+2	252
ϕ_i	complex 10	−2	20
S	complex singlet	−4	2
ψ_p, $p = 1, 2, 3$	left-handed 16_F	+1	—

Indices run from 0 to 9, with Euclidean vector metric δ_{ij} and $\epsilon_{0\ldots9} = +1$. Fix $*\Sigma = +i\Sigma$; its conjugate is the $\overline{126}$. The scalar kinetic convention is

$$\mathcal{L}_{\rm kin} = \frac{1}{2}(D_\mu\Phi_{ij})^*(D^\mu\Phi_{ij}) + \frac{1}{2\,5!}(D_\mu\Sigma_{ijklm})^*(D^\mu\Sigma_{ijklm}) + (D_\mu\phi_i)^*(D^\mu\phi_i) + (\partial_\mu S)^*(\partial^\mu S).$$

The self-dual factor avoids double counting complementary five-index sectors. The physical construction uses the usual gauge-curvature kinetic term and $\mathcal{L} = \mathcal{L}_{\rm gauge} + \mathcal{L}_{\rm kin} + \mathcal{L}_Y - V$. The allowed symbolic Yukawa structures are $\psi\psi\phi$ and $\psi\psi\Sigma^*$, with complex symmetric family matrices and isometrically normalized intertwiners. Their numerical matrices and right-handed-neutrino spectrum have not been selected.

The field-content motivation follows Babu and Khan [1]; PQ/axion context is discussed in [2,3]. The potential below is a separately specified source-derived variant. Published scalar benchmarks, fermion fits, and unification conclusions are not imported as results for it. No novelty claim is made.

2. Complete frozen scalar definition

Write $V = V_2 + V_3 + V_4$. Repeated tensor indices are summed over all ten values. Define

$$N = \frac{1}{5!}\Sigma_{abcde}\Sigma^*_{abcde}, \quad u = \phi_i^*\phi_i, \quad t = \phi_i\phi_i, \quad p_n = \mathrm{Tr}(\Phi^n),\ n = 2,3,4,$$

$$K_{ij} = \frac{1}{4!}\Sigma_{iabcd}\Sigma^*_{jabcd}, \qquad T_{ij} = \frac{1}{4!}\Sigma_{iabcd}\Sigma_{jabcd},$$

$$L_{\Phi\Sigma} = \frac{1}{2\,3!}\Phi_{ij}\Phi_{kl}\Sigma_{ikabc}\Sigma^*_{jlabc}, \qquad E = \frac{1}{2!\,2!\,3!}\Sigma_{ijlmn}\Sigma_{ijpqr}\Sigma^*_{lmpqr}\phi_n.$$

T is symmetric traceless. For increasing tuples I, J of length $5-k$, let $P_k[I,J] = \sum_{K\in\binom{\{0,\ldots,9\}}{k}} \Sigma_{KI}\Sigma_{KJ}$ and $Q_k = \sum_{I,J}|P_k[I,J]|^2$, for $k = 0,1,2$. Antisymmetric components with repeated indices vanish. If A, B are disjoint, $\varepsilon(A,B)$ is the parity of their concatenation relative to increasing order; otherwise it is zero. The fourth real five-form quartic is

$$X_{131} = \sum_{I,J\in\binom{\{0,\ldots,9\}}{4}} \sum_{\substack{A\subset I,\ |A|=3\\ B\subset J,\ |B|=1}} \varepsilon(A, I\setminus A)\varepsilon(B, J\setminus B)\varepsilon(A,B)\varepsilon(I\setminus A, J\setminus B)$$
$$\times\, P_1[I,J]\,\overline{P_1[\mathrm{sort}(A\cup B), \mathrm{sort}((I\setminus A)\cup(J\setminus B))]}.$$

The potential in these normalizations is

$$\begin{aligned}
V_2 &= m_\Phi^2 p_2 + m_\Sigma^2 N + m_\phi^2 u + m_S^2|S|^2,\\
V_3 &= \mu_\Phi p_3 + \mu_{\Phi\phi}\Phi_{ij}\phi_i^*\phi_j + [z_6 t S^* + \text{h.c.}],\\
V_4 &= \lambda_{\Phi 1}p_2^2 + \lambda_{\Phi 2}p_4 + \lambda_{\Phi\Sigma 1}p_2 N + \lambda_{\Phi\Sigma 2}L_{\Phi\Sigma}\\
&\quad + \lambda_{\Phi\phi 1}p_2 u + \lambda_{\Phi\phi 2}\phi_i^*\Phi_{ij}\Phi_{jk}\phi_k + \lambda_{\Phi S}p_2|S|^2\\
&\quad + \lambda_{\Sigma 1}Q_0 + \lambda_{\Sigma 2}Q_1 + \lambda_{\Sigma 3}Q_2 + \lambda_{\Sigma 4}X_{131}\\
&\quad + \lambda_{\Sigma\phi 1}Nu + \lambda_{\Sigma\phi 2}\phi_i^*K_{ij}\phi_j + \lambda_{\phi 1}u^2 + \lambda_{\phi 2}|t|^2\\
&\quad + \lambda_{\Sigma S}N|S|^2 + \lambda_{\phi S}u|S|^2 + \lambda_S|S|^4\\
&\quad + [z_4\Phi_{ij}T_{ij}S + z_K\Phi_{ij}\phi_i\phi_j S^* + z_\eta E + z_D T_{ij}\phi_i\phi_j + \text{h.c.}].
\end{aligned}$$

All m^2, μ, λ coefficients are real; $z_6, z_4, z_K, z_\eta, z_D$ are complex. The masses have dimension two, the cubics dimension one, and the quartics dimension zero. Hermitian conjugation includes the coefficient. The shorthand $\mu_{\Phi\phi}, \lambda_{\phi 1,2}, \lambda_{\phi S}, z_\eta$ corresponds to repository names `muPhiPhi`, `lambdaPhiVector1,2`, `lambdaVectorS`, and `zEta`.

There are 29 Hermitian coefficient groups: four quadratic, three cubic, and 22 quartic. Their 24 real and five complex coefficients give 34 real directions before rephasings. Independent D_5 character multiplicities give 34 complex singlet slots before Hermitian pairing; explicit contractions attain that upper bound. The generic scalar phase quotient leaves three relative coupling phases. This basis result fixes the action, not a viable vacuum.

3. Scalar results: a failed tuning and a later positive point

The candidate high-scale background is

$$\langle\Phi\rangle = \frac{\omega}{\sqrt{60}}\,\mathrm{diag}(-2\mathbf{1}_6, 3\mathbf{1}_4), \quad \langle\Sigma\rangle = \frac{\sigma}{4\sqrt{2}}\bigwedge_{k=0}^{4}(e_{2k+1}+ie_{2k}), \quad \langle S\rangle = \frac{v}{\sqrt{2}}, \quad \langle\phi\rangle = 0.$$

The first direction has the intended Pati–Salam/discrete-parity stabilizer; the combined gauge orbit has 33 broken and 12 unbroken directions, consistent with the SM Lie algebra. Global topology and the PQ quotient are separate questions. Tadpoles and Hessians are derived from the same frozen potential.

At the recorded Stage 1 point, $\omega = \sqrt{60}$ in calculation units and $\sigma/\omega = v/\omega = 0.1$. The exact-decimal free coefficients and tadpole-eliminated masses are specified in `STAGE1_POINT.json` and its replay program. All 35 SM-irrep multiplicity blocks were evaluated. The real Hessian has rank 294 and nullity 34: 33 gauge directions plus one PQ direction. The remaining eigenvalues are positive in the canonical kinetic metric; the smallest recorded physical mass squared is approximately 0.08656. This is local positivity modulo symmetry, not global boundedness or a phenomenologically fitted scale. This point has no light Higgs doublet.

The distinct Stage 2 point changes $\mathrm{Re}\,z_6, \mathrm{Re}\,z_K, \mathrm{Re}\,z_\eta$ to 0.3 in the stored dimensionless convention and tunes $m_\phi^2/\omega^2 \simeq -0.1337864654$ to the exact algebraic root recorded in `STAGE2_POINT.json`. It has rank 290 and nullity 38, including four real coordinates of one massless complex SM Higgs doublet. All physical heavy quadratic modes remain positive. This statement alone is insufficient for local stability along the tuned flat direction.

Decisive additional test

For a canonically normalized neutral field h, $H^\dagger H = h^2/2$, expand around the tuned point with physical heavy coordinates X_A:

$$V = V_0 + a_{\mathrm{direct}}h^4 + h^2 j_A X_A + \tfrac{1}{2}X_A\mathcal{H}_{AB}X_B + \cdots .$$

Minimization on the positive heavy subspace gives $X = -\mathcal{H}^{+}jh^2 + \cdots$, and hence

$$\lambda_{\mathrm{eff}} = 4\left(a_{\mathrm{direct}} - \tfrac{1}{2}j^T\mathcal{H}^{+}j\right), \qquad V_{\mathrm{eff}} = \lambda_{\mathrm{eff}}(H^\dagger H)^2 + \cdots .$$

The recorded parent-action evaluation gives

Quantity in $(H^\dagger H)^2$ normalization	Value
Fixed-heavy direct quartic	+0.19910
Heavy relaxation subtraction	56.68893
Relaxed effective quartic	**−56.48984**

The negative sign makes the Stage 2 point a failed local scalar minimum at tree level. It does not invalidate the separate Stage 1 result or prove that every parameter point fails. A direct parent-potential expansion along the heavy-field valley reproduces the quartic; heavy-coordinate shear and complex Higgs-phase controls preserve it. These extraction routes share the frozen tensor implementation. No global-minimum or loop-corrected stabilization claim is made.

A distinct positive-quartic point

The later recorded point in `canonical_so10_positive_higgs/POINT.json` retains the Stage 1 VEVs but uses $z_6/\omega = z_K = 0.03$, $z_\eta = 0.3$, $\lambda_{\Phi\phi 1} = -0.002$ and an exact algebraic tuning $m_\phi^2/\omega^2 \simeq 0.007302047861$. Its real Hessian has rank 290 and nullity 38, with positive heavy modes. The direct quartic is 0.19845235; heavy relaxation subtracts 0.09359411, leaving

$$\boxed{\lambda_{\mathrm{eff}} = +0.10485824.}$$

The point's SHA-256 is `476645438ea8a7ada42fea90e80174b772d45ce3e729d013d134e9848950c816`. Direct-parent and valley-polynomial routes agree with shared tensor inputs. This remains a local tree-level result. Subsequent calculation at this SAME point gives a physical coherent scalar channel with $|a_0| = 0.76809032 > 1/2$ in the fixed-input high-energy tree limit [7]. Its vector-exchange currents vanish in that channel. The point therefore fails this perturbativity screen; positive local masses and quartic did not establish a controlled benchmark. This failure does not exclude the construction. The new exploratory point in Section 6 has different inputs and must be assessed separately.

4. Gauge matching: a measured mismatch, an incomplete verdict

The recorded direct one-loop calculation matches the full parent to the SM in a common $\overline{\mathrm{MS}}$ background-field Feynman-gauge convention, with GUT-normalized g_1. It includes 290 heavy real scalar directions and 33 massive vector/Goldstone/ghost sets from the later positive-quartic point, not the earlier failed tuning. The threshold ledger binds that point's hash. This point subsequently fails the scalar perturbativity screen. These residuals have not been recomputed for the new exploratory point in Section 6.

Define $\alpha_i = g_i^2/(4\pi)$ and one-loop threshold functions λ_i by

$$\alpha_i^{-1}(\mu) = \alpha_{10}^{-1}(\mu) - \frac{\lambda_i(\mu)}{12\pi}.$$

The complete state accounting gives

$$b_{\mathrm{SM}} = \left(\frac{41}{10}, -\frac{19}{6}, -7\right), \quad b_S = \left(\frac{377}{30}, \frac{77}{6}, \frac{79}{6}\right), \quad T_V = (8, 6, 5),$$
$$b_{\mathrm{SM},i} + b_{S,i} - \frac{7}{2}T_{V,i} = -\frac{34}{3} = b_{10}, \qquad \frac{d\lambda_i}{d\ln\mu} = 6(b_{\mathrm{SM},i} - b_{10}).$$

These are exact beta-jump and one-loop matching-scale checks. At $\mu = \omega$, the frozen inverse-coupling inputs are $(59.0100, 29.5800, 8.47458)$ at M_Z. The one-loop equations are

$$\alpha_i^{-1}(M_Z) = \frac{4\pi}{g_{10}^2} - \frac{\lambda_i^S + \lambda_i^V(g_{10})}{12\pi} + \frac{b_i^{\mathrm{SM}}}{2\pi}\ln\frac{\omega}{M_Z}.$$

They have no common perturbative solution for the frozen inputs. The recorded least-squares residuals, in inverse-coupling units and channel order $(1, 2, 3)$, are:

Completed approximation	r_1	r_2	r_3	$\max\lvert r_i\rvert$
One-loop running and thresholds	0.36615	−1.05980	0.69389	1.05980
Two-loop SM running, one-loop thresholds	0.26091	−0.74548	0.48479	0.74548

The latter uses the SM running framework [5] and frozen reference boundaries from [6], including top, bottom, and tau insertions with their one-loop evolution. A separate inverse-coupling evolution with a different adaptive integrator reproduces the residual to better than 2×10^{-8} under the frozen equations. Matching-scale variation $\mu/\omega \in [0.5, 2]$ leaves maximum residuals from 0.74172 to 0.74931.

What the missing calculation must supply

The exact two-loop RG logarithmic template has been derived, but its finite model-dependent heavy constants have not. The published general matching framework [4] provides methodology and a restricted comparator, not the finite answer for these mixed, nondegenerate fields and multiple VEVs. The complete Yukawa-dependent threshold also requires numerical family matrices and right-handed-neutrino masses.

Small scale variation is not a rigorous bound on unknown finite constants. Setting them to zero or freely tuning them would not compute a prediction. Consequently the formal gauge verdict remains unresolved: neither unification success nor exclusion has been established. The diagnostic fitted $g_{10} \simeq 0.5231$ and $\omega \simeq 1.64 \times 10^{15}\,\mathrm{GeV}$ are not physical benchmark parameters. No replacement spectrum or flavor fit is introduced in this note.

5. Symmetry and quantum evidence: scope matters

A separately named covariant physical gauge/scalar candidate satisfies exact classical local gauge identities through factorized representation checks. The recorded primary and tensor-functor replay compare the 328-real scalar representation action and all 45 generator directions. The usual $\delta(Dx) = -g_{10}\rho(\omega)Dx$ identity, invariant kinetic metric, curvature identity, and invariant potential establish the classical statement. This qualification is inactive; it does not replace a promoted component AST or complete a quantum calculation.

Earlier literal component-action failures are retained. One preserved two-field witness has nonzero gauge variation

$$\delta\mathcal{L}_{\text{old}} = \frac{ig_{10}}{\sqrt{2}}(\partial_\mu C^4)(Y\partial^\mu X - X\partial^\mu Y),$$

with nonzero ghost Euler derivative. The later covariant candidate fixes the kinetic structure by derivation. A failed encoded candidate is evidence about that representation of the action, not a theorem against the gauge group.

Separately, the recorded one-loop parent scalar-pole projection has exact rank $4 + 4 + 26 = 34$ and zero out-of-basis residual in its tested operator space. Hermiticity and PQ selection-rule audits also pass within that scope. This bounded radiative-closure result is not a complete anomaly or quantum consistency theorem; it does not evaluate finite two-loop thresholds.

In the gauge-fixing/source investigation, a preserved normalization candidate misses a selected auxiliary–antighost–ghost cubic pole. At one recorded real-component witness, in common $\hbar g_{10}\xi^2/(32\pi^2\epsilon)$ units, the coefficients are

$$G_{\text{actual}} = \frac{48\sqrt{10}}{25}, \quad G_{\text{candidate}} = \frac{8\sqrt{10}}{5}, \quad G_{\text{actual}} - G_{\text{candidate}} = \frac{8\sqrt{10}}{25} \neq 0.$$

Later calculations retain this failed candidate while deriving actual additional source, auxiliary and ghost coefficients and checking selected local Ward/Slavnov relations. These are scoped operator results. They do not establish the full source-free Slavnov identity or a quantum-equivalence theorem.

The latest selected background test verifies that a specified broken-antighost operation commutes with the actual local residual-gauge transformation on all 1334 classical field/source/background columns. In its declared polynomial ring, a 642-column Ward/Euler calculation proves uniqueness of a background extension at coupling orders zero and one, conditional on locality and inherited covariance of the actual logarithmic UV pole. A nonlocal transverse counterexample shows why locality is essential. Subsequent tests verify weighted principal invertibility on all 643 fields and construct one explicit local covariant reference operator, recovering all 1597 flat entries. This supplies a local-symbol member, without identifying a unique historical completion or adopting a physical scheme. The actual product of heat-damped propagators still lacks a proved UV/subtraction correspondence; the full background logarithmic identity remains conditional. Regulated insertions, products/contacts, measure and infrared limits remain open. The formal ledger is still at 34 of 39 items with the quantum gate blocked.

Gauge anomaly and PQ structure

For the three left-handed 16_F multiplets, the local four-dimensional perturbative gauge-anomaly tensor and linear gauge traces vanish. A weight polynomial route with conjugacy/polarization and a Clifford-matrix route evaluate the established representation criterion [9], including all 16,215 symmetric cubic slots. This does not construct a regulated matter determinant or settle global anomalies, bundles or the full quantum measure.

For the declared PQ charges the convention $N = 2\sum_f q_f T_c(f)$ and $E = 2\sum_f q_f Q_f^2$ gives

$$N = 12, \qquad E = 32, \qquad E/N = 8/3.$$

The common Spin(10) center identifies an ordinary-field PQ quarter-turn; nonzero gauge-singlet S fixes the vacuum-orbit period to $\alpha = \pi/2$. The conventional QCD potential would consequently give $N_{\text{DW}} = 3$, *conditional* on realistic nonzero quark masses/QCD susceptibility, this ordinary orbit and no additional identification or explicit PQ breaking [3]. The electroweak-unbroken reference does not supply that QCD vacuum. This is neither a physical axion mass/abundance prediction nor a full global completion.

6. A new scalar witness and its quantum-control warning

The different rational fixture in `canonical_so10_coherent_joint_witness/POINT.json` uses the same field content, action and VEV shape. Its frozen reduced inputs are $p_1 = 1/100$, $p_2 = 1/50$, $x = 3/25$, $s = 1/1000$, $z = 1/500$, $D = 1/50$, $t = 23/1250$, $\Delta = 0$, $\eta = 1/10000$; the dependent coefficients and larger simple weak root are fixed by the recorded action maps. It is an exploratory fixture, not an adopted physical benchmark. Do not combine its results with the old point's spectrum, thresholds or loops.

Exact specialization of the inherited 12 spectral types (35 blocks, 328 real fields) gives 290 positive directions and 38 symmetry/tuning zeros. The actual weak kernel and all positive neutral relaxation modes give $\lambda_{\rm eff} \simeq 0.1074374161 > 0$ at tree level. This uses a new point-specific light direction, not a transported earlier quartic.

Complete classical coverage at this point

The complete symmetric scalar pair space has 53,956 states. In the contact convention of [7], the recorded structural Gram and block-norm bounds give

$$\|a_0^{\rm contact}\| \le \frac{35439836953}{240000000000\pi} \simeq 0.47003544 < \frac{1}{2}.$$

This is a conservative upper bound, not a largest eigenvalue or complete gauged scattering amplitude. Massless-gauge exchange requires its own infrared-safe treatment; no gauge coupling is silently set to zero.

With $X = \operatorname{Tr}\Phi^2$, $n = N_\Sigma$, $u = \phi^\dagger\phi$, $v = |S|^2$, global invariant inequalities and explicit completion of the complex mixed terms give a rational positive-definite four-norm lower matrix. Its exact minors and a separate rational solver yield, for all 328 real fields,

$$V_4 \ge c(X + n + 2u + 2v)^2, \qquad c = \frac{82440401071117326624733394 9}{17205798939483426705901101207 50} \simeq 4.79143115 \times 10^{-4} > 0.$$

This is global quartic positivity, rather than sampled positive rays. It ensures growth at large fields for this potential. It does not prove that the chosen stationary vacuum is deepest or has a sufficient tunneling lifetime. The old failed points and closed parameter families remain failed in their qualified scopes; these certificates do not transfer to other inputs.

A complete selected one-loop pole coefficient is large

At the same fixture, $Q^2 = \omega^2 = 60$ and the declared $\overline{\rm MS}$ stationary-branch expansion, the isolated singlet has tree squared mass $6/5$. Scalar seagulls and momentum-dependent bubbles, including the physical PQ/Higgs zero modes, give a one-particle-irreducible contribution $-1.36799138 - 0.23557167i$. The required vacuum-source insertion adds $+3.03144395$, so [8]

$$\delta m^2_{S,\rm pole} = 1.66345257 - 0.23557167i, \qquad \frac{|\delta m^2_{S,\rm pole}|}{m^2_{S,\rm tree}} \simeq 1.40004.$$

Field-content and projector checks justify this particular one-loop component. Direct singlet gauge/Yukawa vertices vanish here. The tree singlet plane is isolated and the relevant other tadpole multipliers vanish. This does not make every quantum-vacuum component complete. An exact local RG/log check retains generated Φ–S and Σ–S portals [10] and cancels the scale derivative, but does not reduce the large correction. A small first-order correction is unsupported in this expansion. No reliable physical pole mass/width, resummed solution or universal scheme failure is inferred. Classical feasibility and controlled quantum viability remain distinct.

7. Evidence and reproduction map

This note summarizes recorded evidence at source commit

`22709b33f20484efe16eb4c3592d690c8e0650f3`.

The source scientific authority remains `project_state.json`, version 113, unchanged by publication. The original Version 2 note and two base source archives remain preserved. The Version 2 *source update* overlays those archives; the reproduction guide specifies extraction, deleted paths and separate historical-policy/external-control namespaces. Manifests authenticate bytes without performing scientific acceptance. Original Git history is not included; historical custody exit 2 with seven unresolved references is not PASS.

Evidence paths below are relative to the assembled source tree. Each new row includes its prospective `RECORD.md`, result, actual validation receipt, frozen inputs, sources and complete successful/failed captures.

Claim	Evidence locator under `calculations/`
Scalar action	`canonical_so10_scalar_reconstruction/RESULT_V2.md`
Original masses	`canonical_so10_scalar_benchmark/RESULT.md`
Failed / positive quartics	`canonical_so10_light_higgs_quartic/RESULT.md` and `canonical_so10_positive_higgs/RESULT.md`
Old point contact failure	`canonical_so10_radial_input_control/RESULT.md`
New point and local tests	`canonical_so10_coherent_joint_witness/RESULT.md`
Whole contact bound	`canonical_so10_complete_contact_bound/RESULT.md`
Global quartic bound	`canonical_so10_finite_full_bfb/RESULT.md`
Singlet pole warning	`canonical_so10_singlet_pole_control/RESULT.md`
Pole RG/log check	`canonical_so10_singlet_pole_rg_check/RESULT.md`
Gauge residuals	`canonical_so10_direct_gauge_matching/RESULT.md` and `canonical_so10_direct_gauge_next_order/RESULT.md`
Missing finite boundary	`canonical_so10_direct_two_loop_heavy_threshold/RESULT.md`
Gauge / PQ traces	`canonical_so10_local_gauge_anomaly/RESULT.md` and `canonical_so10_pq_anomaly_screen/RESULT.md`
Conditional background	`canonical_so10_ub12_background_locality/RESULT.md`

The scalar-pole closure record is under `tools/two_loop_qft_compiler_v1/LAYER5B_M05_M08_RESULT.md`. The preserved classical-action and auxiliary-normalization records are in its `m08_independent_replay_package_v1/` subdirectory: `A10_PHYSICAL_ACTION_RECONCILIATION_RESULT.md` and `A10_E5_AUXILIARY_VERTEX_RESULT.md`. The larger source archive retains intermediate exclusions, corrections and bounded source/ghost calculations; archive inclusion does not promote every historical claim. The completed weighted-reference and local-witness records are also included; work after the frozen checkpoint is excluded.

What publication validation does

The package verifier checks every released file and archive member. The publication receipt separately authenticates the selected scientific receipts and their immutable inputs, distinguishes historical policy versions and records executed short checks. No expensive Hessian, contact, singlet-pole or full quantum calculation is repeated for this release. Original paths in external control scripts may require explicit relocation in a disposable environment; flat export is not an automatic historical audit replay.

8. Remaining obligations and AI assistance

No physically viable GUT or experimental agreement is established. The new point has a full classical quartic bound, but the preferred vacuum, tunneling, loop control, hierarchy/naturalness and complete gauged scattering remain open. Full quantum restoration/measure/contact/IR obligations, finite heavy thresholds, realistic matter/neutrino inputs, complete axion cosmology and proton-decay phenomenology also remain open. The old gauge residuals are diagnostics at their own frozen inputs and do not predict the new point.

Useful discussion questions are whether the new point's large singlet correction admits a justified, controlled physical-input expansion, and what model-specific locality and finite-threshold calculations are needed before extending the quantum and gauge conclusions. Missing contributions are not assumed to rescue any failed result.

LLMs assisted derivations, code, verification and writing. Scientific direction and acceptance belong to the human author. The records demonstrate bounded symbolic/numerical calculation, structural reductions and cross-checking; they also preserve failed interpolation, assertion, normalization and verification attempts. Corrected agreement cannot validate a shared physical assumption. Shared action/representation inputs, inherited proof caches, some tensor/arithmetic code, Python/SymPy and the same analyst limit independence. Selected work uses NumPy, SciPy, mpmath, CVXPY/Clarabel and Julia. Multiple AI routes are not independent scientists. No external expert verification, whole-model proof-assistant certification or general reliability claim is made.

References

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